

Atomic Layer Etching of SiO₂ for Nanoscale Semiconductor Devices: A Review

Received 22 December, 2023; accepted 28 December, 2023

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ABSTRACT

In this paper, atomic layer etching (ALE) processes for SiO₂ are reviewed and categorized into two distinct group of anisotropic and isotropic ALE processes. Anisotropic ALE typically involves the fluorination of the silicon dioxide (SiO₂) surface by fluorocarbon deposition during surface modification, followed by the removal of fluorinated layers by relatively low-energy ions. The impacts of the precursor, ion energy, selectivity, and chamber wall conditions on anisotropic ALE processes are reviewed. Isotropic ALE involves the conversion of SiO₂ surfaces into a fluorinated layer or ammonium salt. This layer is subsequently eliminated through various chemical reactions, such as sublimation, fluorination, and ligand exchange. The mechanisms of etching in isotropic ALE are reviewed and classified into two subcategories of thermally isotropic and plasma-assisted isotropic ALE.

Keywords: Anisotropic etching, Atomic layer etching, Isotropic etching, Silicon oxide

1. Introduction

Silicon dioxide (SiO₂) has been applied as an insulator in various nanoscale semiconductor devices, playing a crucial role in the semiconductor industry for the past 50 years [1,2]. SiO₂ exhibits exceptional insulating performance, a bulk resistivity of 10¹⁵ Ω cm, and a dielectric breakdown strength of 10⁷ V cm⁻¹. Additionally, it is cost effective and easy to manufacture and demonstrates excellent compatibility with silicon bulk [3].

Plasma etching is a crucial and essential processing technique in semiconductor device fabrication. Its application to next-generation semiconductor devices is becoming increasingly challenging as the critical dimension (CD) of semiconductors decreases to 10 nm level [4–7]. With decreasing CD and the adoption of three-dimensional structures, conventional reactive-ion etching processes are facing limitations in thickness controllability, etch selectivity, and surface roughness at the nanoscale [6–8]. Consequently, atomic layer etching (ALE) processes are under active development, offering atomic-level precision in layer removal, minimized surface roughness, and exceptional uniformity [4–15].

ALE is a cyclic process that facilitates the atomic-level removal of various layers through a modification step involving radicals or molecules, followed by a removal step utilizing ions or chemical reactions as shown in Fig. 1. ALE processes can provide precise thickness control, excellent surface roughness, and high uniformity at both atomic and nanometer scales [16–21]. A typical ALE process comprises four steps. Initially, the precursor chemisorbs onto the substrate surface through a chemical reaction, which may or may not be self-limiting. The second step involves purging to remove any physically adsorbed reactants. In the third step, modified surface layers are removed, forming volatile etch products via energetic ions or chemical reactions.

It is classified as anisotropic ALE when the products are removed faster in one direction by directional energetic ions and as isotropic ALE when removal occurs uniformly in all directions. The final step involves purging the chamber with inert gases, similar to the second step. This sequence is then repeated in subsequent cycles.

In this review, recent advancements in the ALE of SiO₂ are categorized into anisotropic and isotropic ALE processes [8,22–43]. For anisotropic SiO₂ ALE, we summarize the effects of precursors, ion energy, selectivity, and chamber wall conditions. Isotropic SiO₂ ALE processes are reviewed in terms of various etching mechanisms such as

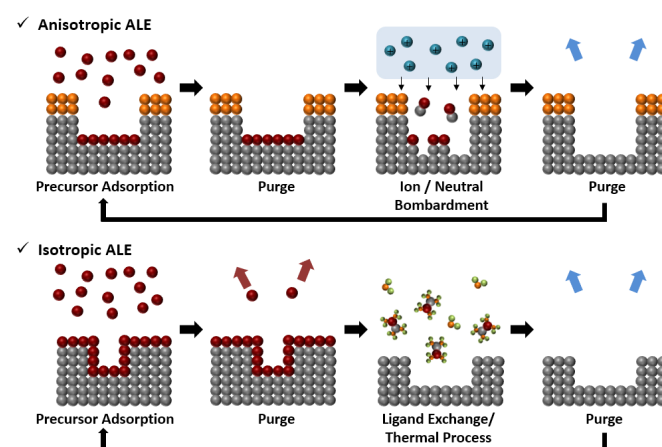


Figure 1. Comparative schematic of anisotropic and isotropic ALE processes for SiO₂.

Table I. Summary of fluorocarbon precursors utilized in anisotropic ALE of SiO₂.

Precursor chemistries for fluorination	Removal	Process temp. (°C)	Etching rate (Å/cycle)	Ref.
C ₄ F ₈ /Ar plasma	Ar plasma	10	2.5	[24]
CHF ₃ /Ar plasma	Ar plasma		3.5	
C ₄ F ₆ /Ar plasma	Ar plasma	−10	13	[40]
C ₄ F ₈ /Ar plasma	Ar plasma		2.6	[31]
C ₄ F ₈ /H ₂ /Ar plasma	Ar plasma	10	1.6	
C ₃ H ₃ F ₃ /Ar plasma	Ar plasma		1.2	
CHF ₃ plasma	O ₂ plasma	RT	6.8	[30]
	Ar plasma		4.0	
CF ₃ I/Ar plasma	O ₂ plasma	40	9.3	[41]
C ₄ F ₈ plasma	Ar plasma	RT	5.8	[8]
CHF ₃ plasma	Ar plasma		4.1	
n-C ₃ F ₇ OCH ₃ plasma	O ₂ plasma		2.1	
n-C ₃ F ₇ OCH ₃ plasma	Ar plasma	RT	2.1	[43]
i-C ₃ F ₇ OCH ₃ plasma			1.8	
CF ₃ CF ₂ CF ₂ CH ₂ OH plasma			5.2	

surface modification and the formation of volatile products, encompassing conversion, fluorination, and ligand exchange.

2. Anisotropic SiO₂ ALE

2.1. Fluorocarbon precursors

For anisotropic SiO₂ ALE, the SiO₂ surface undergoes fluorination through the deposition of a fluorocarbon layer, using a range of fluorocarbon-based precursors, as summarized in Table I. Various precursors, including C₄F₈, C₄F₆, CHF₃, C₃H₃F₃, CF₃I, and C₄H₃F₇O isomers, are employed in SiO₂ ALE. The chemical composition of the deposited fluorocarbon film, influenced by the precursor type, significantly impacts the etching characteristics. Studies have compared the ALE processes using C₄F₈/Ar and CHF₃/Ar plasmas, focusing on fluorocarbon and hydrofluorocarbon precursors [24]. These studies analyze the composition of the fluorocarbon film and the etching rate in relation to the C₄F₈ and CHF₃ precursors, establishing a correlation between the F/C ratio in the fluorocarbon film and the SiO₂ etching rate with different precursors.

The addition of hydrogen atoms is known to reduce the fluorine radicals in fluorocarbon plasmas, whereas oxygen atoms enhance it [44–46]. SiO₂ ALE processes using C₄F₈, C₄F₈/H₂, and C₃H₃F₃ plasmas were investigated to determine the impact of hydrogen addition [31]. The fluorocarbon films generated by C₄F₈/H₂ and C₃H₃F₃ plasmas exhibited a lower F/C ratio than those produced by C₄F₈ plasma. Consequently, the SiO₂ etching rate was higher in the C₄F₈ plasma

Table II. Ion energy windows in anisotropic ALE of SiO₂.

Precursor chemistries for fluorination	Removal	Ion energy in the removal step (Bias voltage)	Process temperature (°C)	Etching rate (Å/cycle)	Ref.
CHF ₃ plasma	Ar plasma	0–50V	−40–20	9.0–11.0	[33]
C ₄ F ₈ plasma beam	Ar ion beam	30–200V	RT	1.9	[27]
C ₄ F ₆ /Ar plasma	Ar plasma	10–100V	−10	14.2	[40]
C ₄ F ₈ plasma	Ar plasma	30–90V	20	5.8	[8]
CHF ₃ plasma	Ar plasma			4.1	
C ₃ F ₇ OCH ₃ plasma	O ₂ plasma			2.1	
n-C ₃ F ₇ OCH ₃ plasma	Ar plasma	10–80V	RT	2.1	[43]
i-C ₃ F ₇ OCH ₃ plasma				1.8	
CF ₃ CF ₂ CF ₂ CH ₂ OH plasma				5.2	

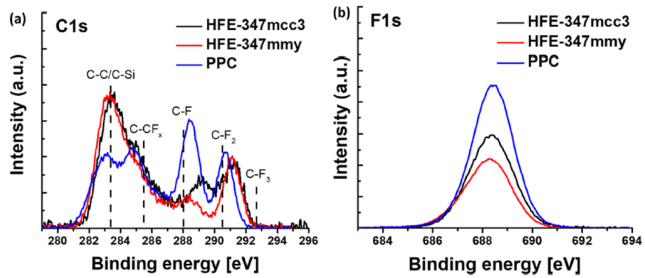


Figure 2. XPS spectra of fluorocarbon films on SiO₂ surfaces: (a) C 1s and (b) F 1s, derived from n-C₃F₇OCH₃ (HFE-347mcc3), i-C₃F₇OCH₃ (HFE-347mmy), and CF₃CF₂CF₂CH₂OH (PPC) plasmas. Reproduced with permission from [43], Copyright 2023, American Chemical Society.

compared to the C₄F₈/H₂ and C₃H₃F₃ plasmas, due to a higher F/C ratio of fluorocarbons on the SiO₂ surface. Furthermore, the SiO₂ ALE process utilizing C₄F₈, CHF₃, and C₃F₇OCH₃ plasmas has been examined to assess the effects of both hydrogen and oxygen additions [8]. The fluorocarbon film formed by the C₃F₇OCH₃ plasma exhibited the lowest F/C ratio compared to those produced by the C₄F₈ and CHF₃ plasmas.

The SiO₂ ALE process utilizing C₄H₃F₇O isomer plasmas was explored to elucidate the impact of chemical structure [43]. The composition of the resulting fluorocarbon film is dictated by the chemical bonding structure of the C₄H₃F₇O isomers, leading to variations in the F/C ratios as shown in Fig. 2. The F/C ratios of fluorocarbons generated by n-C₃F₇OCH₃ and i-C₃F₇OCH₃ plasmas are lower compared to those formed by CF₃CF₂CF₂CH₂OH plasma, which can be attributed to the CH₃ radicals stemming from the presence of -OCH₃ in the molecule. Correspondingly, the etching rate of SiO₂ is highest in the CF₃CF₂CF₂CH₂OH plasma and lowest in the i-C₃F₇OCH₃ plasma, mirroring the F/C ratio of the fluorocarbon films. It is observed that the addition of hydrogen to the plasma reduces the F/C ratio of fluorocarbon films and subsequently diminishes the etching rate of SiO₂.

2.2. Ion energy: Threshold energy and ALE window

In the anisotropic SiO₂ ALE process, the parameters defining the ALE window include the precursor used in the fluorination step, the type of gas employed in the removal step, and the ion energy, as summarized in Table II. The threshold physical sputtering energies identified are 50 eV for SiO₂ [24], 20 eV for Si₃N₄ [23], and 20 eV for Si [24]. Investigations into the physical sputtering of SiO₂, based on the Ar bias voltage, revealed sputtering at ion energies above 65 eV [27]. Surface modification allows for etching with lower ion energy and enables self-limiting removal of the modified layer [7]. In the ALE window region, selective removal of the modified layer from the bulk material

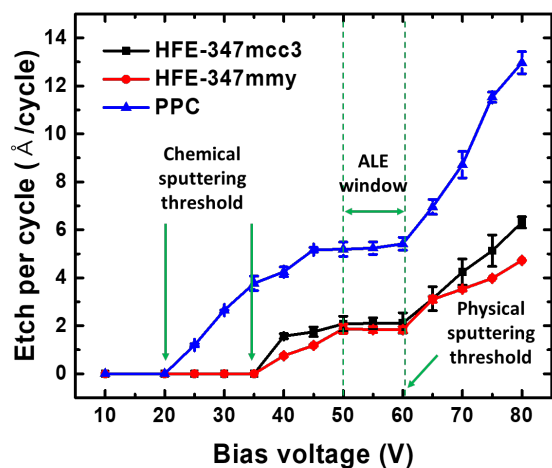


Figure 3. SiO₂ etching rates correlated with bias voltage in etching step, highlighting incomplete etching, ALE window, and physical sputtering regions based on ion energy. Reproduced with permission from [43], Copyright 2023, American Chemical Society.

occurs. Three distinct regions are discerned based on ion energy: the incomplete etching region, the ALE window region, and the physical sputtering region, as shown in Fig. 3 [43]. The removal of the modified layer is not complete at low ion energies, while sputtering of the underlying material occurs at higher ion energies.

The SiO₂ ALE window region has been reported with varying parameters. A previous study reported a window region of 50–60 V with 15 W source power of Ar plasma [8]. Another study identified an SiO₂ ALE window region with a bias voltage of approximately 15 eV at 1 kW of Ar source plasma power, highlighting a very low ion energy range for the ALE window due to the high source power [42]. These findings imply that interactions between chemical structure, ion energy, and process conditions determine the efficacy and characteristics of SiO₂ ALE processes.

2.3. Selectivity

The etching selectivity of SiO₂/Si and SiO₂/Si₃N₄ can be enhanced through various approaches, including the choice of precursors, control of fluorocarbon film thickness, ion energy management, etch step time adjustment, and selective deposition techniques, as outlined in Table III. The selectivity in anisotropic SiO₂ ALE is influenced by the thickness of the fluorocarbon film, which in turn is determined by the

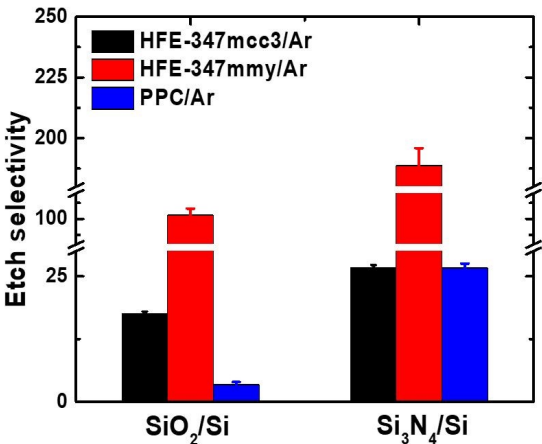


Figure 4. Precursor-dependent etching selectivity analysis for SiO₂/Si and Si₃N₄/Si in fluorination step. Reproduced with permission from [43], Copyright 2023, American Chemical Society.

choice of precursor [23]. Improvements in SiO₂/Si and SiO₂/Si₃N₄ etch selectivity have been achieved using C₄F₈, C₄F₈/H₂, and C₃H₃F₃ plasmas by studying the impact of hydrogen addition [31]. The etching selectivity of SiO₂/Si and SiO₂/Si₃N₄ were notably improved using the C₃H₃F₃ precursor, attributed to the reduction of fluorine concentration in the fluorocarbon film due to hydrogen. Additionally, the enhancement of SiO₂/Si etch selectivity has been discussed using C₄H₃F₇O isomer plasma, as shown in Fig. 4 [43]. A higher presence of Si-C bonds was observed in the fluorocarbon films generated by the i-C₃F₇OCH₃ plasma, compared to those from n-C₃F₇OCH₃ and CF₃CF₂CF₂CH₂OH plasmas. These Si-C bonds act as inhibitors for Si etching, thereby decreasing the etching rate of Si and enhancing the SiO₂/Si etch selectivity. This evidence suggests that carefully selecting precursors can lead to high etch selectivity in anisotropic ALE processes. Moreover, selective functionalization of the SiN_x surface with benzaldehyde has been shown to improve the etch selectivity of SiO₂/SiN_x, as shown in Fig. 5 [38]. Benzaldehyde selectively deposits on SiN_x surfaces featuring -NH_x (x = 1, 2) groups, but not on SiO₂ surfaces with -OH groups. This selective deposition of benzaldehyde on SiN_x surfaces fosters the formation of a hydrofluorocarbon film, which serves as a barrier against the etching of SiN_x. The implication of these findings is that through strategic precursor selection and surface functionalization, the etching selectivity for different material combinations in anisotropic ALE processes can be effectively manipulated and optimized.

Table III. Studies of etch rate selectivity in anisotropic ALE of SiO₂.

Selectivity of material	Selectivity	Improving etch selectivity method	Precursor chemistries for fluorination	Removal	Ref.
SiO ₂ /Si ₃ N ₄	0.2-15.0	Precursor selection, Fluorocarbon film thickness, Ion energy, Etching step time	C ₄ F ₈ /Ar plasma CHF ₃ /Ar plasma	Ar plasma	[24]
SiO ₂ /Si ₃ N ₄ SiO ₂ /Si	>7.0 >10.0	Precursor selection	C ₄ F ₈ /Ar plasma C ₄ F ₈ /H ₂ /Ar plasma C ₃ H ₃ F ₃ /Ar plasma	Ar plasma	[31]
SiO ₂ /Si	2.6-17.5	Precursor selection	C ₄ F ₈ plasma CHF ₃ plasma C ₃ F ₇ OCH ₃ plasma	Ar plasma O ₂ plasma	[8]
SiO ₂ /Si	17.5 102.8 3.4	Precursor selection	n-C ₃ F ₇ OCH ₃ plasma i-C ₃ F ₇ OCH ₃ plasma CF ₃ CF ₂ CF ₂ CH ₂ OH plasma	Ar plasma	[43]
SiO ₂ /Si ₃ N ₄	2.1-4.5	Selective deposition (benzaldehyde)	C ₄ F ₈ /Ar plasma	Ar plasma	[38]

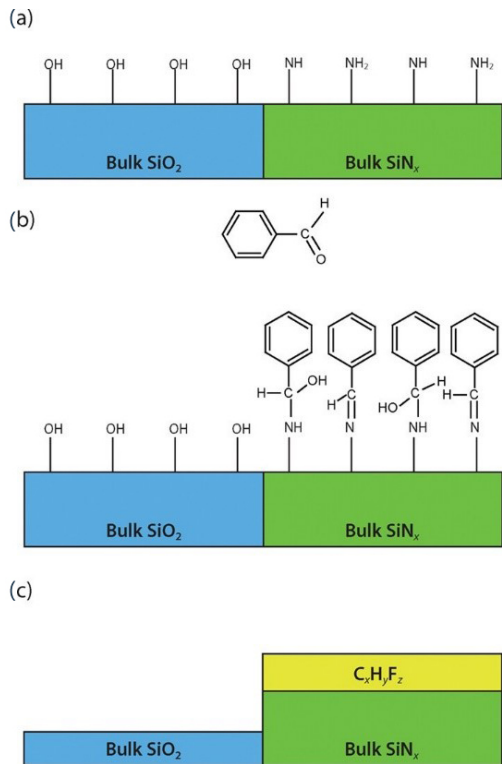


Figure 5. Selective ALE process of SiO₂ over SiN_x: (a) Initial functional groups on plasma-deposited surfaces; (b) selective functionalization of SiN_x surface with benzaldehyde; (c) aromatic aldehyde aiding graphitic hydrofluorocarbon film formation on SiN_x surface during ALE, inhibiting SiN_x etching. Reproduced with permission from [38], Copyright 2021, AIP Publishing.

2.4. Chamber wall effect

In the anisotropic SiO₂ ALE process, the etching rate is influenced by the fluorocarbon film left on chamber walls, as summarized in Table IV. Achieving a consistent etching rate is crucial in ALE and the fluorocarbon film on the chamber walls plays a significant role in the repeatability of the etching rate. Variations in the SiO₂ etching rate across multiple ALE cycles have been reported using *in situ* ellipsometry [26]. It was observed that a fluorocarbon film deposited on the chamber wall leads to an increase in the etching rate as the cycle repetition increased. The impacts of the quartz window temperature and a fluorocarbon film on the chamber wall on the etching rate was reported as shown in Fig. 6 [22]. Research focusing on mitigating the chamber wall effect has been conducted using O₂ plasma to maintain a constant etching rate [29]. The application of O₂ plasma effectively prevented the buildup of a fluorocarbon film on the chamber walls, thereby preserving the initial state of the chamber.

Table IV. Studies on impact of chamber wall conditions in anisotropic ALE of SiO₂.

Precursor chemistries for fluorination	Removal	Method of removal chamber wall effect	Etching rate (Å/cycle)	Ref.
C ₄ F ₈ /Ar plasma	O ₂ plasma	O ₂ plasma	5.6–11.4	[29]
C ₄ F ₈ /Ar plasma	Ar plasma	Chamber cleaning with O ₂ plasma, Chamber wall heating	3.0	[22]

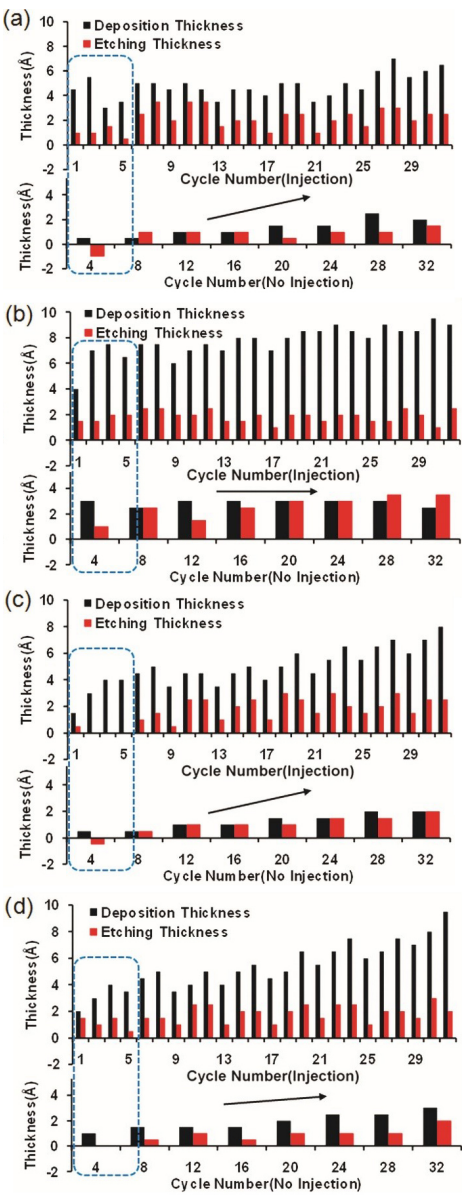


Figure 6. Analysis of deposition and etching thickness under four conditions: (a) Cold/clean, (b) cold/with film, (c) hot/clean, and (d) hot/with film. Reproduced with permission from [22], Copyright 2016, AIP Publishing.

3. Isotropic SiO₂ ALE

3.1. Thermal isotropic ALE

In isotropic SiO₂ ALE processes, the SiO₂ surface undergoes modification through either thermal reactions or plasma-assisted methods. The etching mechanisms for thermal isotropic SiO₂ ALE are summarized in Table V. Here, the SiO₂ surfaces are modified by conversion to Al₂O₃ or by formation of ammonium salt, and then the modified layers are removed through various chemical reactions including sublimation, fluorination, and ligand exchange. These etching mechanisms are categorized into two groups: conversion of SiO₂ into Al₂O₃ or ammonium fluorosilicate (AFS) during the modification step. An example of the isotropic SiO₂ ALE process using triethylaluminum (TMA) and hydrogen fluoride (HF) precursors was reported [25]. In this process, SiO₂ is converted to Al₂O₃ utilizing the TMA precursor, as indicated in Eq. (1), which represents the reaction for modification

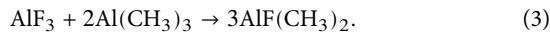
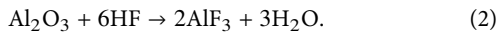
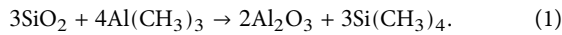
Table V. Studies on thermal isotropic ALE of SiO₂.

Etching mechanism	1 st step	2 nd step	3 rd step	Etching rate (Å/cycle)	Ref.
Conversion: SiO ₂ → Al ₂ O ₃	TMA (300 °C)	HF (300 °C)	TMA (300 °C)	0.07–0.27	[25]
Fluorination: Al ₂ O ₃ → AlF ₃					
Ligand exchange: AlF ₃ → AlF(CH ₃) ₂	TMA (350 °C)	HF (350 °C)	TMA (300 °C)	0.35–1.50	[32]
Conversion: SiO ₂ → (NH ₄) ₂ SiF ₆	HF (20 °C)	NH ₃ (20 °C)	Heating (140 °C)	9	[42]
Heating: (NH ₄) ₂ SiF ₆ → SiF ₄ + 2NH ₃ + 2HF					

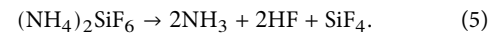
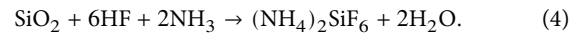
Table VI. Studies on plasma-assisted isotropic ALE of SiO₂.

Etching mechanism	1 st step	2 nd step	3 rd step	Etching rate (Å/cycle)	Ref.
Conversion: SiO ₂ → (NH ₄) ₂ SiF ₆	CF ₄ /NH ₃ plasma	Heating (160 °C)	–	27–70	[35]
Heating: (NH ₄) ₂ SiF ₆ → SiF ₄ + 2NH ₃ + 2HF	NF ₃ /NH ₃ plasma (20 °C)				
	NF ₃ /H ₂ plasma (20 °C)	NH ₃ (20 °C)	Heating (150 °C)	75	[39]

step. Subsequently, Al₂O₃ undergoes fluorination to AlF₃ by HF, as shown in Eq. (2). Finally, AlF₃ is removed via a ligand exchange reaction forming volatile AlF(CH₃)₂ with TMA, as shown in Eq. (3), which represents the reaction for removal step. These reactions occur spontaneously at temperatures as high as 300 °C without plasma assistance.

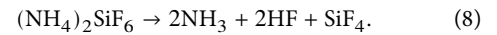


Isotropic SiO₂ ALE process using HF and NH₃ gas has been also reported [42]. The AFS layer is formed by the reaction of NH₃ molecules with the adsorbed HF on SiO₂ surface, as shown in Eq. (4). In the removal step, the AFS layer decomposes at temperatures above 140 °C, forming volatile reaction products such as NH₃, HF, and SiF₄, as shown in Eq. (5).



3.2. Plasma-assisted isotropic ALE

The etching mechanisms of plasma-assisted isotropic SiO₂ ALE are comprehensively summarized in Table VI. A specific isotropic SiO₂ ALE process employing CF₄/NH₃ or NF₃/NH₃ plasma has been examined. In this process, SiO₂ is converted to AFS in CF₄/NH₃ or NF₃/NH₃ plasma as shown in Eqs. (6) and (7). Subsequently, the AFS layer decomposes at temperatures above 160 °C, forming volatile substances such as NH₃, HF, and SiF₄, as shown in Eq. (8). The self-limiting property of AFS formation was confirmed to be dependent on the plasma duration, with the etching rate escalating from 2.7 to 7.0 nm/cycle based on the gas ratio, as shown in Fig. 7. The etching rate observed in the NF₃/NH₃ plasma was approximately threefold higher than that in the CF₄/NH₃ plasma. This discrepancy is attributed to the different bonding energies of F (~506 kJ/mol) and N-F (~239 kJ/mol). NF₃ is more prone to dissociation than CF₄ under similar conditions, leading to the generation of a greater number of fluorine atoms in the NF₃/NH₃ plasma compared to the CF₄/NH₃ plasma. This results in more extensive AFS formation and consequently, a higher etching rate. This step is crucial as it ensures the complete removal of the modified layer, enabling the process to proceed to the next cycle effectively. The distinction in etching rates between the two plasma types underscores the importance of gas selection and plasma conditions in optimizing the isotropic ALE process for SiO₂.



4. Conclusions

In this review, we categorized recent research on the SiO₂ ALE process into anisotropic and isotropic processes. For anisotropic SiO₂ ALE processes, the effects of the precursor, ion energy, selectivity, and chamber wall conditions were examined. The choice of precursor influenced the F/C ratio in the film deposited on the SiO₂ surface, which subsequently affected the etching rate and selectivity. In the anisotropic ALE process, changes in ion energy affected the etching rate, and a consistent etching rate was observed within the defined ALE window. For isotropic SiO₂ ALE, we elucidated two types of mechanisms are summarized. SiO₂ surface into a fluorinated layer or

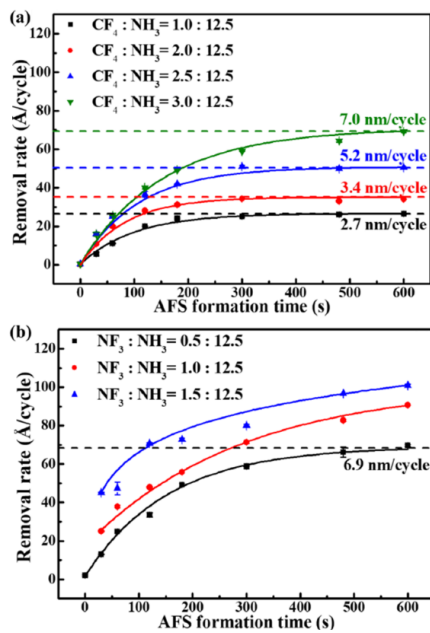


Figure 7. AFS formation time-dependent removal rates with (a) CF₄/NH₃ plasma and (b) NF₃/NH₃ plasma. Reproduced with permission from [35], Copyright 2020, AIP Publishing.

ammonium salt, followed by removal through various chemical reactions, including sublimation, fluorination, and ligand exchange. Given the increasing complexity and three-dimensional nature of semiconductor device integration, the necessity for both anisotropic and isotropic ALE processes is evident. The selection of specific SiO₂ ALE mechanisms should be tailored to the device architecture. Future research on the nuances of the SiO₂ ALE mechanism is essential to further enhance the precision and efficiency of these processes in semiconductor fabrication.

Acknowledgments

This work was supported by the Technology Innovation Program (RS-2022-00155706), funded by the Ministry of Trade, Industry & Energy (MOTIE, Korea).

Conflict of interest

The authors declare no conflicts of interest.

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References

- [1] J. Coburn, H. F. Winters, and T. J. Chuang, *J. Appl. Phys.* 48, 3532 (1977).
- [2] H. F. Winters and J. W. Coburn, *Surf. Sci. Rep.* 14, 162 (1992).
- [3] F. Palumbo, C. Wen, S. Lombardo, S. Pazos, F. Aguirre, M. Eizenberg, F. Hui, and M. Lanza, *Adv. Funct. Mater.* 30, 1900657 (2020).
- [4] V. M. Donnelly and A. Kornblit, *J. Vac. Sci. Technol. A* 31, 050825 (2013).
- [5] T. Faraz, F. Roozeboom, H. C. M. Knoop, and W. M. M. Kessels, *ECS J. Solid State Sci. Technol.* 4, N5023 (2015).
- [6] K. J. Kanarik, T. Lill, E. A. Hudson, S. Sriraman, S. Tan, J. Marks, V. Vahedi, and R. A. Gottscho, *J. Vac. Sci. Technol. A* 33, 020802 (2015).
- [7] G. Oehrlein, D. Metzler, and C. Li, *ECS J. Solid State Sci. Technol.* 4, N5041 (2015).
- [8] Y. Kim, S. Lee, Y. Cho, S. Kim, and H. Chae, *J. Vac. Sci. Technol. A* 38, 022606 (2020).
- [9] J. Kim, H. Kang, Y. Kim, M. Jeon, and H. Chae, *Plasma Processes Polym.* e2300216 (2024).
- [10] S. K. Sung, T. Y. Kim, E. S. Cho, H. J. Cho, B. Y. Choi, C. W. Oh, B. K. Cho, C. H. Lee, and D. Park, *IEEE Trans. Nanotechnol.* 5, 174 (2006).
- [11] H. Yun, T. Kim, C. B. Shin, C. -K. Kim, J. -H. Min, and S. H. Moon, *Korean J. Chem. Eng.* 24, 670 (2007).
- [12] N. R. Johnson and S. M. George, *ACS Appl. Mater. Interfaces* 9, 34435 (2017).
- [13] J. Kim, D. Shim, Y. Kim, and H. Chae, *J. Vac. Sci. Technol. A* 40, 032603 (2022).
- [14] Y. Lee, Y. Kim, J. Son, and H. Chae, *J. Vac. Sci. Technol. A* 40, 022602 (2022).
- [15] D. Shim, J. Kim, Y. Kim, and H. Chae, *J. Vac. Sci. Technol. B* 40, 022208 (2022).
- [16] N. R. Johnson, H. Sun, K. Sharma, and S. M. George, *J. Vac. Sci. Technol. A* 34, 050603 (2016).
- [17] Y. Lee and S. M. George, *Chem. Mater.* 29, 8202 (2017).
- [18] A. M. Cano, A. E. Marquardt, J. W. DuMont, and S. M. George, *J. Phys. Chem. C* 123, 10346 (2019).
- [19] S. M. George, *Acc. Chem. Res.* 53, 1151 (2020).
- [20] Y. Kim, H. Kang, H. Ha, M. Choi, M. Jeon, S. M. Cho, and H. Chae, *Plasma Processes Polym.* e2300161 (2023).
- [21] Y. Kim, H. Kang, H. Ha, C. Kim, S. Cho, and H. Chae, *Appl. Surf. Sci.* 627, 157309 (2023).
- [22] M. Kawakami, D. Metzler, C. Li, and G. S. Oehrlein, *J. Vac. Sci. Technol. A* 34, 040603 (2016).
- [23] C. Li, D. Metzler, C. S. Lai, E. A. Hudson, and G. S. Oehrlein, *J. Vac. Sci. Technol. A* 34, 041307 (2016).
- [24] D. Metzler, C. Li, S. Engelmann, R. L. Bruce, E. A. Joseph, and G. S. Oehrlein, *J. Vac. Sci. Technol. A* 34, 01B101 (2016).
- [25] J. W. DuMont, A. E. Marquardt, A. M. Cano, and S. M. George, *ACS Appl. Mater. Interfaces* 9, 10296 (2017).
- [26] R. J. Gasvoda, A. W. van de Steeg, R. Bhowmick, E. A. Hudson, and S. Agarwal, *ACS Appl. Mater. Interfaces* 9, 31067 (2017).
- [27] S. S. Kaler, Q. Lou, V. M. Donnelly, and D. J. Economou, *J. Phys. D: Appl. Phys.* 50, 234001 (2017).
- [28] D. Metzler, C. Li, C. S. Lai, E. A. Hudson, and G. S. Oehrlein, *J. Phys. D: Appl. Phys.* 50, 254006 (2017).
- [29] T. Tsutsumi, H. Kondo, M. Hori, M. Zaitzu, A. Kobayashi, T. Nozawa, and N. Kobayashi, *J. Vac. Sci. Technol. A* 35, 01A103 (2017).
- [30] K. Koh, Y. Kim, C. -K. Kim, and H. Chae, *J. Vac. Sci. Technol. A* 36, 01B106 (2018).
- [31] K. Y. Lin, C. Li, S. Engelmann, R. L. Bruce, E. A. Joseph, D. Metzler, and G. S. Oehrlein, *J. Vac. Sci. Technol. A* 36, 040601 (2018).
- [32] R. Rahman, E. C. Mattson, J. P. Klesko, A. Dangerfield, S. Rivillon-Amy, D. C. Smith, D. Hausmann, and Y. J. Chabal, *ACS Appl. Mater. Interfaces* 10, 31784 (2018).
- [33] S. Dallorto, A. Goodyear, M. Cooke, J. E. Szornel, C. Ward, C. Kastl, A. Schwartzberg, I. W. Rangelow, and S. Cabrini, *Plasma Processes Polym.* 16, 1900051 (2019).
- [34] R. J. Gasvoda, Y. G. Verstappen, S. Wang, E. A. Hudson, and S. Agarwal, *J. Vac. Sci. Technol. A* 37, 051003 (2019).
- [35] Y. Cho, Y. Kim, S. Kim, and H. Chae, *J. Vac. Sci. Technol. A* 38, 022604 (2020).
- [36] G. Antoun, T. Tillocher, P. Lefauchaux, J. Faguet, K. Maekawa, and R. Dussart, *Sci. Rep.* 11, 357 (2021).
- [37] A. Fischer, A. Routzahn, S. M. George, and T. Lill, *J. Vac. Sci. Technol. A* 39, 030801 (2021).
- [38] R. J. Gasvoda, Z. Zhang, E. A. Hudson, and S. Agarwal, *J. Vac. Sci. Technol. A* 39, 040401 (2021).
- [39] Y. J. Gill, D. S. Kim, H. S. Gil, K. H. Kim, Y. J. Jang, Y. E. Kim, and G. Y. Yeom, *Plasma Processes Polym.* 18, 2100063 (2021).
- [40] M. Y. Yoon, H. J. Yeom, J. H. Kim, W. Chegal, Y. J. Cho, D. C. Kwon, J. R. Jeong, and H. C. Lee, *Phys. Plasmas* 28, 063504 (2021).
- [41] S. Kim, I. Park, and J. Ahn, *Appl. Surf. Sci.* 589, 153045 (2022).
- [42] N. Miyoshi, H. Kobayashi, K. Shinoda, M. Kurihara, K. Kawamura, Y. Kouzuma, and M. Izawa, *J. Vac. Sci. Technol. A* 40, 012601 (2022).
- [43] Y. Kim, H. Kang, C. Kim, and H. Chae, *ACS Sustainable Chem. Eng.* 11, 6136 (2023).
- [44] C. Mogab, A. Adams, and D. Flamm, *J. Appl. Phys.* 49, 3796 (1978).
- [45] H. Doh, J. Kim, K. Whang, and S. Lee, *J. Vac. Sci. Technol. A* 14, 1088 (1996).
- [46] D. Marra and E. Aydil, *J. Vac. Sci. Technol. A* 15, 2508 (1997).